

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2012 question paper
for the guidance of teachers

9702 PHYSICS

9702/42

Paper 4 (A2 Structured Questions), maximum raw mark 100

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Section A

- 1 (a) force proportional to product of masses and inversely proportional to square of separation (*do not allow square of distance/radius*)
either point masses or separation @ size of masses M1 A1 [2]
- (b) (i) $\omega = 2\pi / (27.3 \times 24 \times 3600)$ or $2\pi / (2.36 \times 10^6)$ M1
 $= 2.66 \times 10^{-6} \text{ rad s}^{-1}$ A0 [1]
- (ii) $GM = r^3 \omega^2$ or $GM = v^2 r$ C1
 $M = (3.84 \times 10^5 \times 10^3)^3 \times (2.66 \times 10^{-6})^2 / (6.67 \times 10^{-11})$ M1
 $= 6.0 \times 10^{24} \text{ kg}$ A0 [2]
(special case: uses $g = GM/r^2$ with $g = 9.81$, $r = 6.4 \times 10^6$ scores max 1 mark)
- (c) (i) grav. force $= (6.0 \times 10^{24}) \times (7.4 \times 10^{22}) \times (6.67 \times 10^{-11}) / (3.84 \times 10^8)^2$ C1
 $= 2.0 \times 10^{20} \text{ N}$ (allow 1 SF) A1 [2]
- (ii) either $\Delta E_p = Fx$ because F constant as $x \ll$ radius of orbit B1
 $\Delta E_p = 2.0 \times 10^{20} \times 4.0 \times 10^{-2}$ C1
 $= 8.0 \times 10^{18} \text{ J}$ (allow 1 SF) A1 [3]
- or $\Delta E_p = GMm/r_1 - GMm/r_2$ C1
Correct substitution B1
 $8.0 \times 10^{18} \text{ J}$ A1
($\Delta E_p = GMm/r_1 + GMm/r_2$ is incorrect physics so 0/3)
- 2 (a) energy $= \frac{1}{2} m \omega^2 a^2$ and $\omega = 2\pi f$ C1
 $= \frac{1}{2} \times 37 \times 10^{-3} \times (2\pi \times 3.5)^2 \times (2.8 \times 10^{-2})^2$ M1
 $= 7.0 \times 10^{-3} \text{ J}$ A0 [2]
(allow $2\pi \times 3.5$ shown as 7π)
- Energy $= \frac{1}{2} m v^2$ and $v = r\omega$ (C1)
Correct substitution (M1)
Energy $= 7.0 \times 10^{-3} \text{ J}$ (A0)
- (b) $E_K = E_p$ C1
 $\frac{1}{2} m \omega^2 (a^2 - x^2) = \frac{1}{2} m \omega^2 x^2$ or E_K or $E_p = 3.5 \text{ mJ}$ C1
 $x = a/\sqrt{2} = 2.8/\sqrt{2}$ or $E_K = \frac{1}{2} m \omega^2 (a^2 - x^2)$ or $E_p = \frac{1}{2} m \omega^2 x^2$ C1
 $= 2.0 \text{ cm}$ A1 [3]
(E_K or $E_p = 7.0 \text{ mJ}$ scores 0/3)
- Allow: $k = 17.9$ (C1)
 $E = \frac{1}{2} k x^2$ (C1)
 $x = 2.0 \text{ cm}$ (A1)

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	(c) (i) graph: horizontal line, y-intercept = 7.0 mJ with end-points of line at +2.8 cm and –2.8 cm	B1	[1]
	(ii) graph: reasonable curve with maximum at (0,7.0) end-points of line at (–2.8, 0) and (+2.8, 0)	B1 B1	[2]
	(iii) graph: inverted version of (ii) with intersections at (–2.0, 3.5) and (+2.0, 3.5) (Allow marks in (iii), but not in (ii), if graphs K & P are not labelled)	M1 A1	[2]
	(d) <u>gravitational potential</u> energy	B1	[1]
3	(a) sum of potential energy and kinetic energy of atoms/molecules/particles reference to random (distribution)	M1 A1	[2]
	(b) (i) as lattice structure is ‘broken’/bonds broken/forces between molecules reduced (not molecules separate) no change in kinetic energy, potential energy increases internal energy increases	B1 M1 A1	[3]
	(ii) <i>either</i> molecules/atoms/particles move faster/ $\langle c^2 \rangle$ is increasing <i>or</i> kinetic energy increases with temperature (increases) no change in potential energy, kinetic energy increases internal energy increases	B1 M1 A1	[3]
4	(a) (i) as r decreases, energy decreases/work got out (due to <u>attraction</u>) so point mass is negatively charged	M1 A1	[2]
	(ii) electric potential energy = charge $\times$ electric potential electric field strength is potential gradient field strength = gradient of potential energy graph/charge	B1 B1 A0	[2]
	(b) tangent drawn at (4.0, 14.5) gradient = 3.6×10^{-24} (for $< \pm 0.3$ allow 2 marks, for $< \pm 0.6$ allow 1 mark) field strength = $(3.6 \times 10^{-24}) / (1.6 \times 10^{-19})$ = $2.3 \times 10^{-5} \text{ V m}^{-1}$ (allow ecf from gradient value) (one point solution for gradient leading to $2.3 \times 10^{-5} \text{ V m}^{-1}$ scores 1 mark only)	B1 A2 A1	[4]

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- 5 (a) (long) straight conductor carrying current of 1 A
current/wire normal to magnetic field
(for flux density 1 T,) force per unit length is 1 N m^{-1} M1
M1 [3]
A1
- (b) (i) (originally) downward force on magnet (due to current) B1
by Newton's third law (allow "N3") M1
upward force on wire A1 [3]
- (ii) $F = BIL$
 $2.4 \times 10^{-3} \times 9.8 = B \times 5.6 \times 6.4 \times 10^{-2}$ C1
 $B = 0.066 \text{ T}$ (need 2 SF) A1 [2]
(*g* missing scores 0/2, but $g = 10$ leading to 0.067 T scores 1/2)
- (c) new reading is $2.4\sqrt{2} \text{ g}$ C1
either changes between $+3.4 \text{ g}$ and -3.4 g
or total change is 6.8 g A1 [2]
- 6 (a) oil drop charged by friction/beta source B1
between parallel metal plates B1
plates are horizontal (1)
adjustable potential difference/field between plates B1
until oil drop is stationary B1
 $mg = q \times V/d$ B1
symbols explained (1)
oil drop viewed through microscope (1)
 m determined from terminal speed of drop (when p.d. is zero) (1)
(any two extras, 1 each) B2 [7]
- (b) $3.2 \times 10^{-19} \text{ C}$ A1 [1]
- 7 (a) minimum energy to remove an electron from the metal/surface B1 [1]
- (b) gradient = 4.17×10^{-15} (allow $4.1 \rightarrow 4.3$) C1
 $h = 4.15 \times 10^{-15} \times 1.6 \times 10^{-19}$ or $h = 4.1 \text{ to } 4.3 \times 10^{-15} \text{ eVs}$ A1
 $= 6.6 \times 10^{-34} \text{ J s}$ A0 [2]
- (c) graph: straight line parallel to given line B1
with intercept at any higher frequency B1 [3]
intercept at between $6.9 \times 10^{14} \text{ Hz}$ and $7.1 \times 10^{14} \text{ Hz}$

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- 8 (a) nuclei having same number of protons/proton (atomic) number
different numbers of neutrons/neutron number
(allow second mark for nucleons/nucleon number/mass number/atomic
mass if made clear that same number of protons/proton number) B1
B1 [2]
- (b) probability of decay per unit time is the decay constant C1
 $\lambda = \ln 2 / t_{1/2}$
 $= 0.693 / (52 \times 24 \times 3600)$
 $= 1.54 \times 10^{-7} \text{ s}^{-1}$ C1
A1 [3]
- (c) (i) $A = A_0 \exp(-\lambda t)$
 $7.4 \times 10^6 = A_0 \exp(-1.54 \times 10^{-7} \times 21 \times 24 \times 3600)$
 $A_0 = 9.8 \times 10^6 \text{ Bq}$
(alternative method uses 21 days as 0.404 half-lives) C1
A1 [2]
- (ii) $A = \lambda N$ and $\text{mass} = N \times 89 / N_A$
 $\text{mass} = (9.8 \times 10^6 \times 89) / (1.54 \times 10^{-7} \times 6.02 \times 10^{23})$
 $= 9.4 \times 10^{-9} \text{ g}$ C1
A1 [2]

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Section B

- 9 (a) e.g. infinite input impedance/resistance
zero output impedance/resistance
infinite (open loop) gain
infinite bandwidth
infinite slew rate
(any four, one mark each) B4 [4]
- (b) graph: square wave M1
180° phase change A1
amplitude 5.0 V A1 [3]
- (c) correct symbol for LED M1
diodes connected correctly between V_{OUT} and earth A1
diodes identified correctly A1 [3]
(special case: if diode symbol, not LED symbol, allow 2nd and 3rd marks to be scored)
- 10 (a) e.g. beam is divergent/obeys inverse square law
absorption (in block)
scattering (of beam in block)
reflection (at boundaries)
(any two sensible suggestions, 1 each) B2 [2]
- (b) (i) $I = I_0 \exp(-\mu x)$ C1
 $I_0/I = \exp(0.27 \times 2.4)$
 $= 1.9$ A1 [2]
- (ii) $I_0/I = \exp(0.27 \times 1.3) \times \exp(3.0 \times 1.1)$ C1
 $= 1.42 \times 27.1$
 $= 38.5$ A1 [2]
- (c) either much greater absorption in bone than in soft tissue B1 [1]
or I_0/I much greater for bone than soft tissue
- 11 (a) (i) loss of (signal) power B1 [1]
- (ii) unwanted power (on signal)
that is random M1
A1 [2]
- (b) for digital, only the 'high' and the 'low' / 1 and 0 are necessary M1
variation between 'highs' and 'lows' caused by noise not required A1 [2]
- (c) attenuation = $10 \lg(P_2 / P_1)$ C1
either $195 = 10 \lg\{2.4 \times 10^3\} / P$
or $-195 = 10 \lg(P / 2.4 \times 10^3)$ C1
 $P = 7.6 \times 10^{-17} \text{ W}$ A1 [3]

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- 12 (a) (i)** modulator B1 [1]
- (ii) serial-to-parallel converter (*accept series-to-parallel converter*) B1 [1]
- (b) (i)** enables one aerial to be used for transmission and receipt of signals A1 [1]
- (ii) all bits for one number arrive at one time B1
bits are sent out one after another B1 [2]